Zürich, Switzerland & Oslo, Norway
Email: *xiangtian.shi@outlook.com*
LinkedIn: *xiangtian-shi*
GitHub: *xitisch*

Xiangtian Shi

Education

2026 – Present **ETH Zürich, Switzerland**
MSc in Robotics, Systems and Control
Current coursework: Model Predictive Control, Robot Dynamics, Planning and Decision Making for Autonomous Robots, Vision Algorithms for Mobile Robotics.

2023 – 2026 **KTH Royal Institute of Technology, Sweden**
BSc in Vehicle Engineering (GPA: 4.9/5.0)
Selected coursework: Automatic Control, Optimization, Numerical Methods, Applied Computer Science, Signals and Mechanical Systems, Electrical Engineering.

Technical Skills

Programming	Python, MATLAB, Arduino (C++)
Software & Tools	PyTorch, Git, ROS 2 (basic), Linux (basic), Solid Edge, Onshape, ANSYS, RocketPy
Engineering	Control systems, modeling and simulation, embedded systems, mechanical design, hardware integration, experimental setup design, data acquisition

Technical Experience and Projects

09.2026 – Present **Real-World Robotics – Robot Arm with Articulated Hand**
Project Course, ETH Zürich, Switzerland
- Working with reinforcement-learning policy training in a Linux and ROS 2 environment.
- Reviewing literature and investigating approaches to long-horizon manipulation using a robot arm equipped with an articulated hand.

06.2026 – Present **Biomimetic Flapping-Wing Robot – Sim-to-Real Research**
Visiting Student Researcher, Dr. Tao Du's Group, IIIS, Tsinghua University, China
- Developed an ESP32-based wireless SBUS control interface for the group's flapping-wing research; integrated and troubleshot experimental hardware.
- Designed mechanical components for motion-capture and force-measurement data acquisition to improve measurement reliability and support sim-to-real validation.
- Investigating custom wing designs, fabrication methods, and material selection for ongoing experimental research.

01.2026 – 06.2026 **PINNs for Optimal Trajectory Planning of Unicycle Robots**
Bachelor's Thesis, KTH, Sweden
- Developed PyTorch frameworks for physics-informed trajectory optimization under kinematic and control constraints.
- Implemented physics-based constraint enforcement and obstacle-avoidance strategies for unicycle trajectory planning.
- Evaluated loss balancing and constraint enforcement across the optimization workflow.

01.2025 – 09.2025 **Rocket Avionics Engineer**
AESIR Student Rocketry Association, KTH, Sweden
- Developed a ballistic-coefficient apogee predictor maintaining $\leq$0.6% error after settling ($\sim$1.5 s post-burnout), vs. up to 4.2% for the drag-table baseline in RocketPy.
- Co-developed a PID airbrake controller for the Freyja rocket with 100 Hz apogee prediction, achieving 0.2% apogee error in nominal RocketPy simulations.

01.2024 – 06.2024 **Solar-Powered Model Vehicle**
Project Course, KTH, Sweden
- Designed and built a vehicle integrating solar power, mechanical components, and electrical subsystems.
- Implemented Arduino-based control and evaluated the integrated system through performance testing.

Work Experience

10.2025 – 06.2026 **Teaching Assistant in Electrical Engineering**
Unit of Mechatronics, KTH, Sweden
- Supported laboratory instruction in circuit theory, signal processing, and mechatronics.
- Evaluated assignments and assisted with course coordination.

05.2023 – 01.2025 **Mathematics Tutor**
MentorNorge AS, Norway
- Delivered individualized mathematics tutoring adapted to each student's learning needs.

10.2022 – 06.2023 **Kindergarten Assistant**
Høvik kindergarten, Norway
- Supported children aged 2–4 in daily routines and structured creative and educational activities.

Languages

Norwegian (Native); Chinese (Native); English (Fluent; IELTS Academic 7.5); Swedish (Fluent; CEFR B2)

Extracurricular Activities

06.2025 **Volunteer Interpreter**
20th I.S.P.O. World Congress, Stockholm, Sweden
- Provided interpretation in English, Swedish, Chinese and Norwegian for a Chinese manufacturer of prosthetic and orthotic devices.

09.2023 – 06.2024 **International Group**
KTH Student Council, Sweden
- Organized social and cultural events supporting integration among international students.

10.2020 – 08.2022 **Competitive Esports**
Professional PUBG Mobile Player
- Won the PUBG Mobile Pro League Western Europe Spring 2021 in a 16-team professional field and competed internationally.

Hobbies and Interests

Violin (10+ years, regional competition winner); Table tennis (club member, regional competitions); Running (Stockholm Half Marathon 2025); Hiking; Skiing